

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

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Experiment No.: 5-1

1. TITLE

Interactive Product Calculator

2. AIM

To design a small web application that receives a product label, quantity, and price, then computes and presents the payable amount interactively in the browser.

3. OBJECTIVE

The objectives of this experiment are:

- To construct an input form for collecting product and pricing details.
 - To connect HTML controls with JavaScript through DOM selection.
 - To convert typed quantities and prices into numerical values for calculation.
 - To respond to both button clicks and field changes with event listeners.
 - To prevent invalid entries from producing an incorrect total.
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4. THEORY

The calculator uses HTML for the interface, CSS for presentation, and JavaScript for the application logic. The browser stores each entered value inside its corresponding form control.

JavaScript accesses these controls through `document.getElementById()`. The `Number()` function converts quantity and price from input text into numeric data. A calculation function then checks the values and multiplies the quantity by the price.

The result field is modified using `textContent`, so the displayed amount changes directly on the current page. Event listeners attached to the button, quantity field, and price field make the interface responsive to user actions.

Calculation rule: Total amount = Number of units × Price of one unit

5. ALGORITHM / PROCEDURE

1. Open an HTML document and place the product, quantity, and price controls inside a container.
2. Assign IDs to the controls so that JavaScript can identify them.
3. Create `calculateTotal()` as the central calculation routine.
4. Fetch the entered product name, quantity, price, and result element.
5. Convert quantity and price into numeric values.

6. Check that a product name has been entered and that quantity and price satisfy the validation rules.
 7. Multiply quantity by price to obtain the total.
 8. Format the value with two decimal places and place it inside the result heading.
 9. Connect the button to calculateTotal() with a click event listener.
 10. Connect input events on quantity and price so changes are reflected immediately.
 11. Load the page in a browser and verify several normal and invalid entries.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Product Calculator</title>
<link rel="stylesheet" href="style.css">
</head>

<body>

<div class="container">

<h1>Interactive Product Calculator</h1>

<label for="productName">Product Name:</label>
<input type="text" id="productName" placeholder="Enter product name">

<label for="quantity">Quantity</label>
<input type="number" id="quantity" min="1" value="1">

<label for="price">Price</label>
<input type="number" id="price" min="0" step="0.01" value="0">

<button id="calculateBtn">Calculate Total</button>

<h2 id="result">Total: ₹0.00</h2>

</div>

<script src="script.js"></script>

</body>
</html>
```

style.css

```
body {
  font-family: Arial, sans-serif;
  background-color: #f2f2f2;
  display: flex;
  justify-content: center;
  align-items: center;
  min-height: 100vh;
}

.container {
  background-color: white;
  padding: 30px;
  width: 350px;
  border-radius: 10px;
  box-shadow: 0 0 10px gray;
}

h1 {
  text-align: center;
}

label {
  display: block;
  margin-top: 15px;
}

input {
  width: 100%;
  padding: 10px;
  margin-top: 5px;
  box-sizing: border-box;
}

button {
  width: 100%;
  padding: 10px;
  margin-top: 20px;
  cursor: pointer;
}

#result {
  text-align: center;
  margin-top: 20px;
}
```

6. PROGRAM (CONTINUED)

script.js

```
function calculateTotal() {

  const productName = document.getElementById("productName").value;
  const quantity = Number(document.getElementById("quantity").value);
  const price = Number(document.getElementById("price").value);

  const result = document.getElementById("result");

  // Validation
  if (productName.trim() === "") {
    result.textContent = "Please enter a product name.";
    return;
  }

  if (quantity <= 0 || price < 0) {
    result.textContent = "Please enter valid quantity and price.";
    return;
  }

  // Calculate total
  const total = quantity * price;

  // Display result
  result.textContent = `Total for ${productName}: ₹${total.toFixed(2)}`;
}

// Get button
const calculateButton = document.getElementById("calculateBtn");

// Add event listener
calculateButton.addEventListener("click", calculateTotal);

// Calculate automatically when quantity changes
document.getElementById("quantity").addEventListener("input", calculateTotal);

// Calculate automatically when price changes
document.getElementById("price").addEventListener("input", calculateTotal);
```

7. CODE EXPLANATION

Code / Feature	Function in the application
calculateTotal()	Reads the current form values, validates them, performs multiplication, and updates the displayed amount.
document.getElementById()	Selects a particular input, button, or result element from the HTML page.
Number()	Converts the quantity and price inputs into numeric data before arithmetic is performed.
trim()	Removes surrounding spaces so an empty product name can be detected correctly.
if (...)	Stops execution and writes a validation message when the supplied values are not acceptable.
quantity * price	Computes the amount owed for the selected quantity.
toFixed(2)	Keeps exactly two digits after the decimal point in the displayed result.
addEventListener()	Connects the calculation routine to button clicks and changes in numeric fields.

8. TEST CASES AND EXECUTION DATA

Test Case	Product	Quantity	Price	Expected Result	Actual Result	Status
TC-01	Pen	15	50.28	Total = Rs. 754.20	Total for pen: Rs. 754.20	Pass
TC-02	Notebook	3	120.00	Total = Rs. 360.00	Total for Notebook: Rs. 360.00	Pass
TC-03	Bottle	4	85.50	Total = Rs. 342.00	Total for Bottle: Rs. 342.00	Pass
TC-04	Marker	8	12.25	Total = Rs. 98.00	Total for Marker: Rs. 98.00	Pass

Additional Validation Test

Test Case	Input	Expected Result	Actual Result	Status
TC-05	Blank product name	Please enter a product name.	Please enter a product name.	Pass
TC-06	Quantity = 0	Please enter valid quantity and price.	Please enter valid quantity and price.	Pass
TC-07	Price = -5	Please enter valid quantity and price.	Please enter valid quantity and price.	Pass

9. OUTPUT

Output 1: Calculator result for the entered product details

The image shows a web-based calculator interface titled "Interactive Product Calculator". It features three input fields: "Product Name:" with the value "pen", "Quantity:" with the value "15", and "Price:" with the value "50.28". Below these fields is a button labeled "Calculate Total". The result is displayed as "Total for pen: ₹754.20".

Field	Value
Product Name	pen
Quantity	15
Price	50.28
Calculate Total	
Total	₹754.20

Observed values: Product = pen, Quantity = 15, Price = 50.28, Total = Rs. 754.20.

10. RESULT

Thus, the Interactive Product Calculator was developed successfully using HTML, CSS, and JavaScript. The final screen shows that the entered quantity of 15 and price of 50.28 are combined to produce Rs. 754.20 for the product pen.

The experiment also confirms that the same calculation routine can respond to edits made in the quantity and price fields and can display validation messages for unacceptable entries.

How this would be submitted

Source Code

```
Q1_Interactive_Product_Calculator/  
├── index.html  
├── style.css  
└── script.js
```